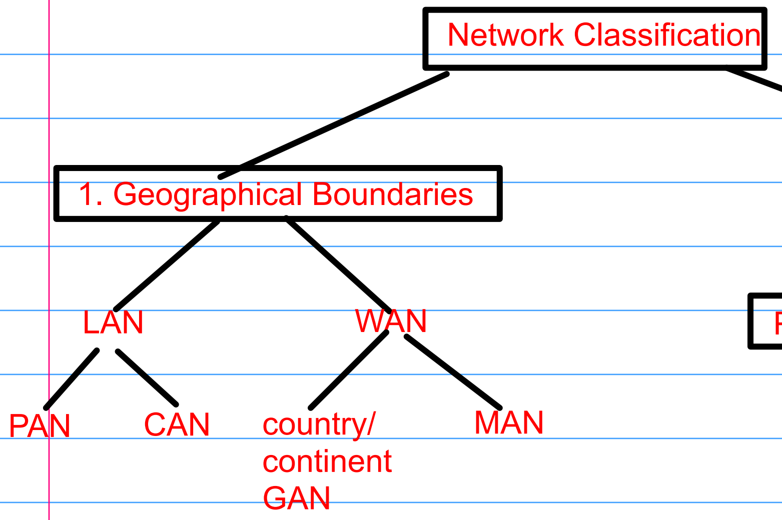
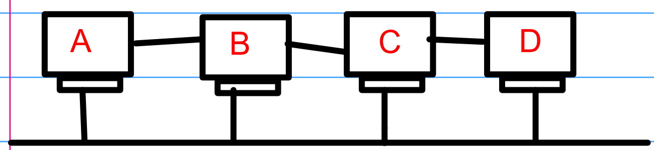


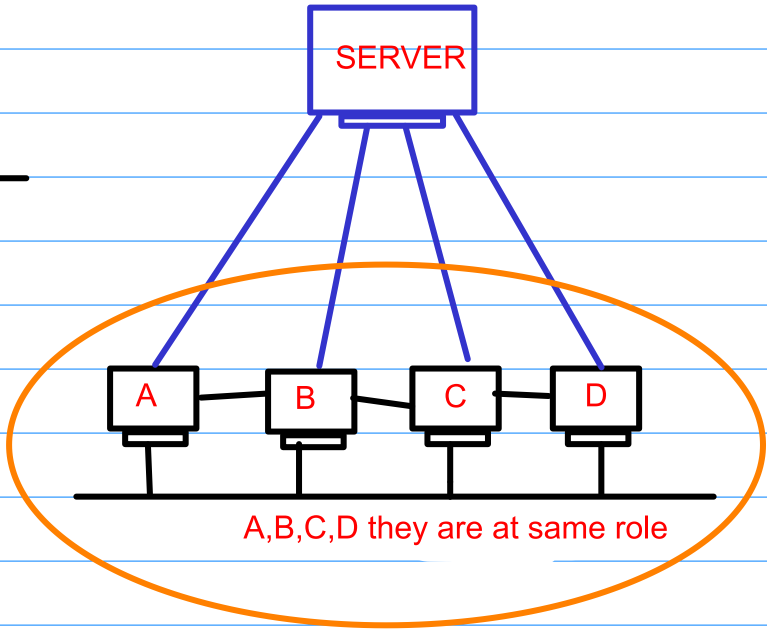
Component Role

1. Peer to Peer Network



Role of all machines is same

2. Client Server Network



Bluetooth-

1. Wireless Personal Area Network(WPAN)
2. radio waves for transmission of data
3. shorter distance
4. FHSS(Frequency Hopping Spread Spectrum) - hopping from one freq to another in a random way
5. One bluetooth network is called piconet
6. group of interconnected piconets is called as scatternet.

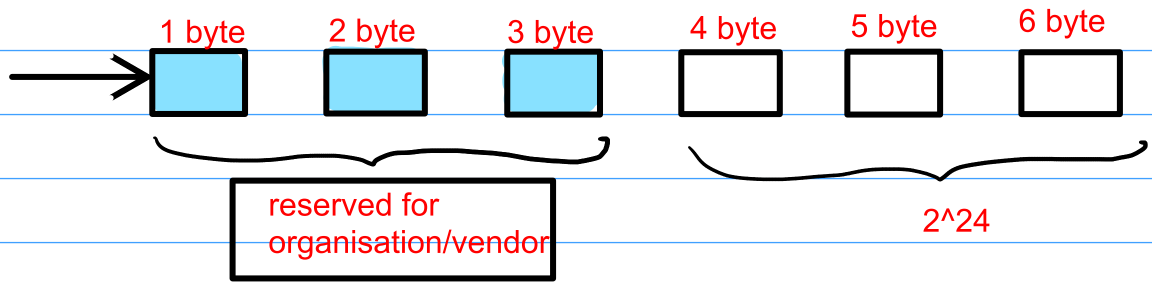
Ethernet Address

NIC/Ethernet card

20-47-47-0-0-1

MAC address = 48 bits(6 bytes)

2^0
2^1
..
..
..
..
..
..
2^47



Dell

ipconfig_all

20-47-47-0-0-1

20-47-47-0-0-2

...
...
...

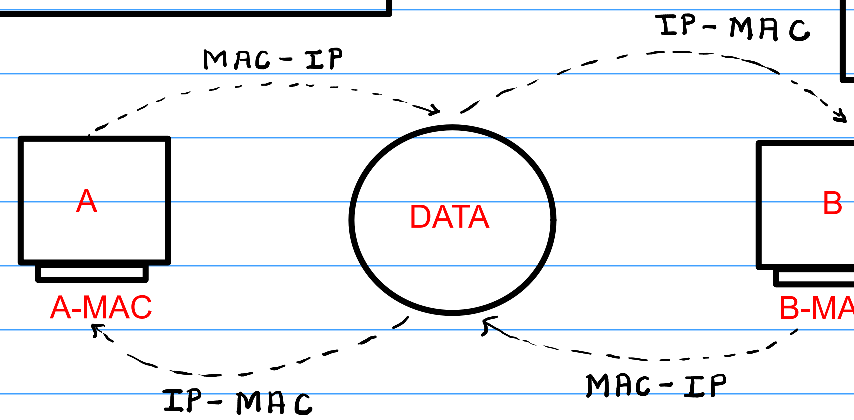
Ethernet Packet(Frame)

Preamble	SFD	Destination MAC	Source MAC	Type	Data and Pad	FCS
7 Bytes	1 Byte	6 Bytes	6 Bytes	2 Bytes	46-1500 Bytes	4 Bytes
START		C	A	http/ https/ IPv4/ IPv6	M1a	2

Address Resolution Protocol(ARP)

MAC - Physical address

IP - Logical Address



ARP is used to convert the MAC address into IP address and IP address into MAC address

MAC address is a unique address(remain same in office as well as at home)

MAC address remains same for a particular machine

IP address is different in office and at home

Physical Structure

Types of Connections

1. Point to Point
(one transmitter and one receiver)

2. Multipoint
(one transmitter and many receivers)

Types of Transmissions

1. unicast - one to one
   eg: phone call between two people

2. Multicast - one to selected group of people
   eg: giving a call to particular group on WhatsApp

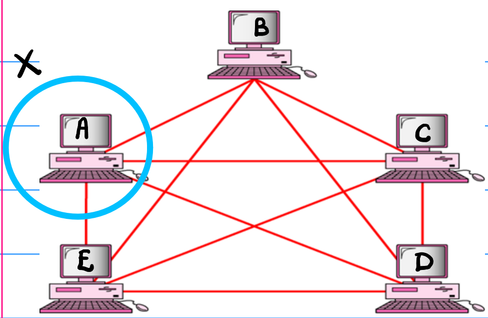
3. Broadcast - one to All
   eg: PM addressing the nation during lockdown at 8pm

Network Topology

- It is the physical layout of all devices,wires(cables) as well as the paths used for data transmission

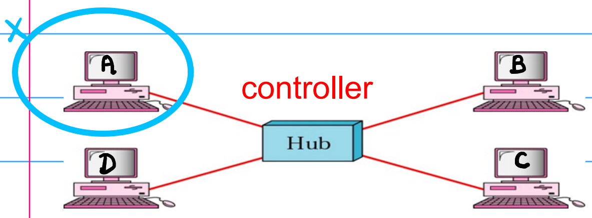
1. Mesh
2. Star
3. Bus
4. Ring

1. Mesh Toplogy



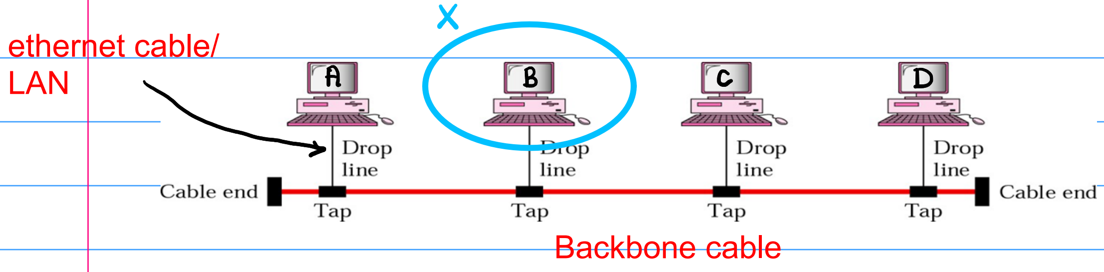
- point to point
- 5 machines (n)
- individual connections = n -1
  = 5 - 1
  = 4
- robust and secured
- very costly
- military comm. or aircraft navigation
- links(n devices) = n(n-1)/2

2. Star Topology



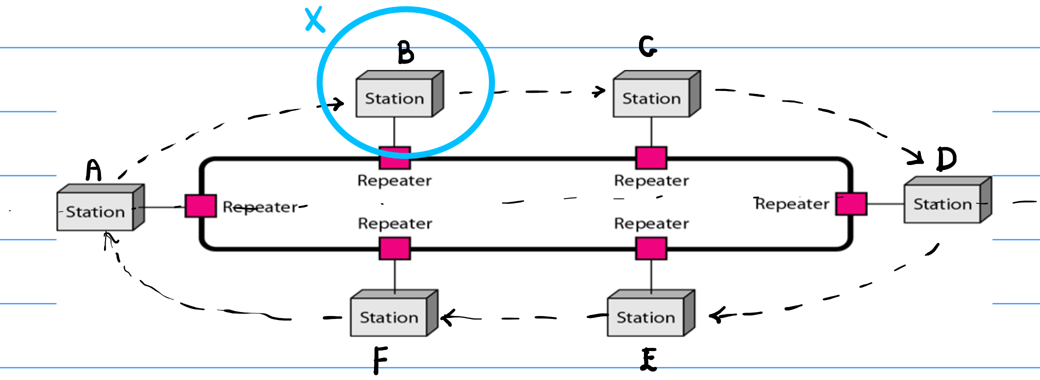
- point to point
- central controller
- controller fails entire system fails
- LAN in offices
- less costly compared to mesh topology

2. Bus Topology



- multipoint
- backbone cable fails entire system fails
- CSMA(Carrier Sense Multiple Access) method

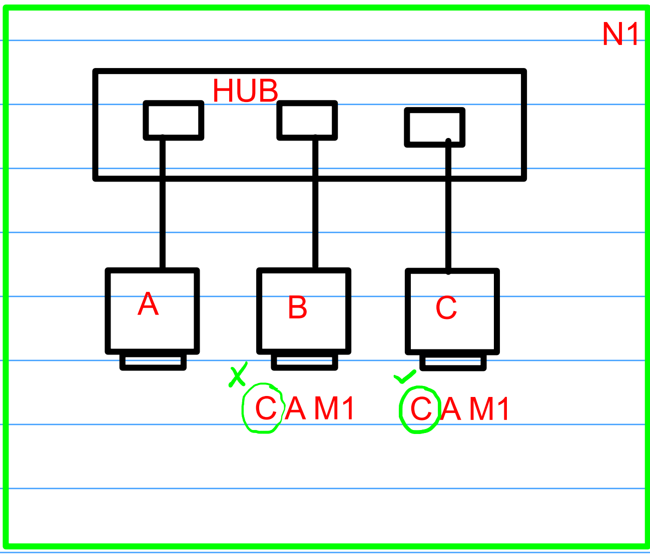
4. Ring Topology



- unidirectional(one direction)
- clockwise direction
- endless continuous loop(single loop)
- no terminating ends
- devices are connected as point to point
- multipoint connection between devices and central controller
- if one device fails then entire network fails
- access method used is token ring method

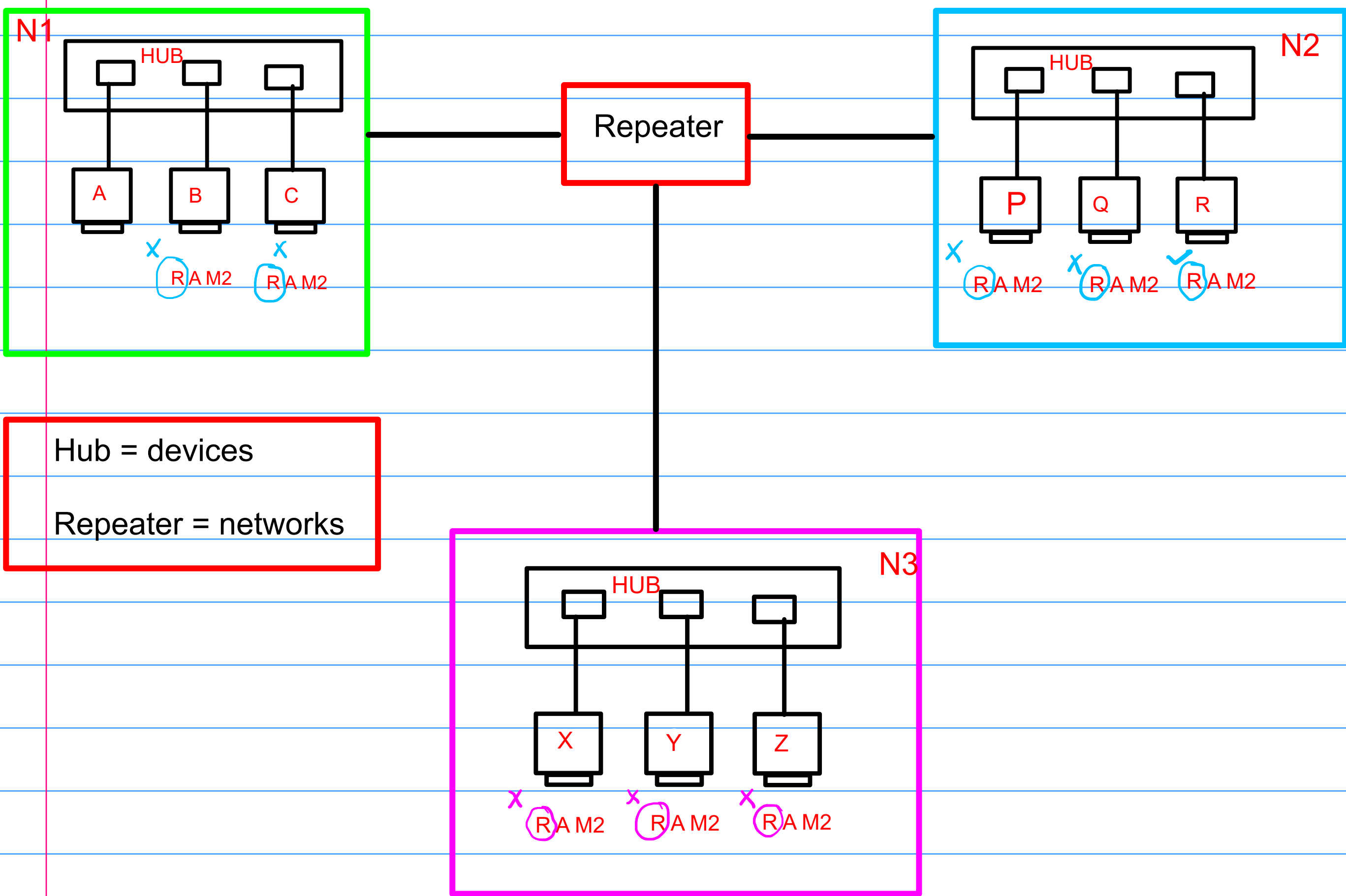
Network Devices -

1. Hub



HUB = devices

2. Repeater

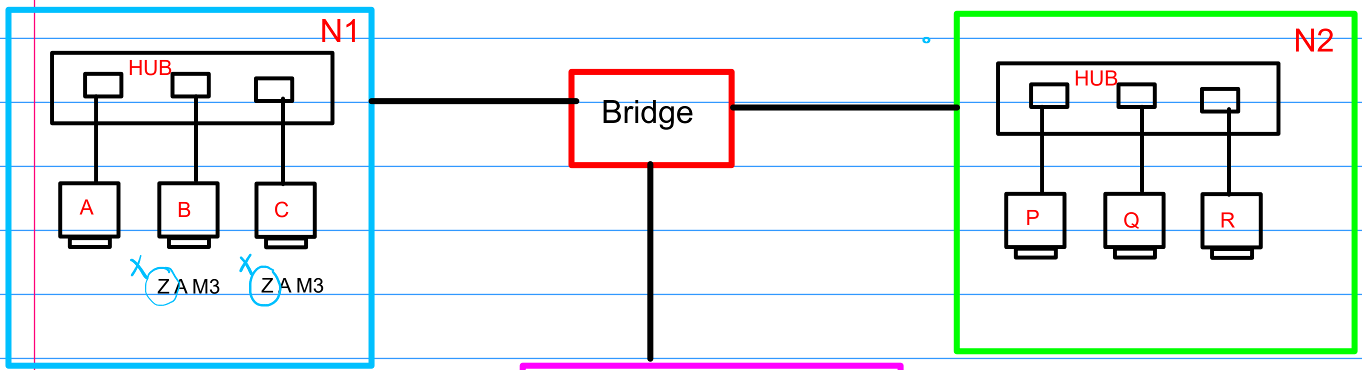


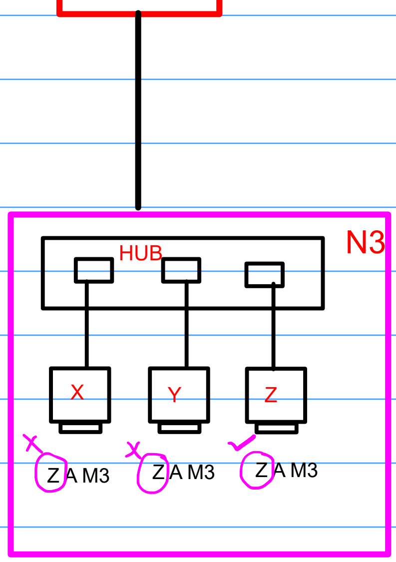
Hub = devices

Repeater = networks

- Repeats the signal or regenerates the network's signal

3. Bridge





Routing Table-

src	IP	MAC	n/w
A	A-IP	A-MAC	N1
...			
P	P-IP	P-MAC	N2
....			
Z	Z-IP	Z-MAC	N3
...			
...			
..			

hub = devices

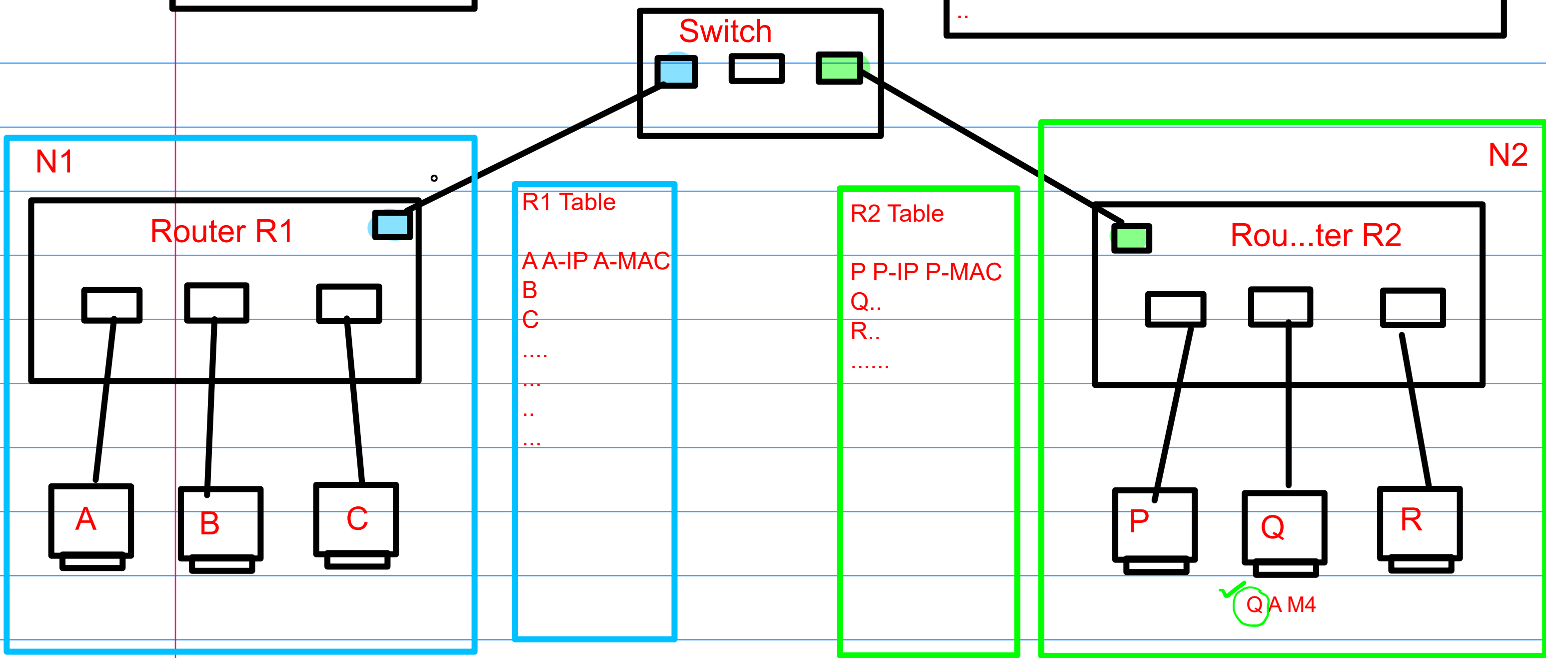
bridge = networks

4. Switch & Router

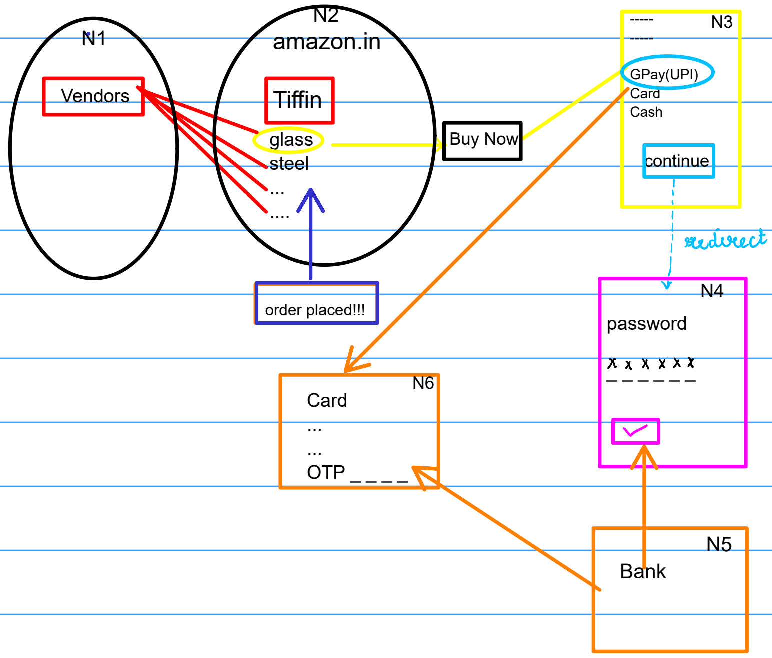
R1 R1-IP R1-MAC N1
R2 R2-IP R2-MAC N2
..
..

A - -
B - -
C - -

P - -
Q - -
R - -



5. Gateway



- device which connects disimilar networks
- operates at highest level of abstraction
- entry exit point for network